

OUSSAMA BATOUCHE

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ABOUT

Applied scientist, currently a Member of Technical Staff at Qutwo, building next-generation AI for the quantum era. I bring a dual background in software engineering and data science, with hands-on experience across machine learning, image analysis, and statistical modelling. Over seven years working with Python and R, with strong skills in deep learning, statistics, and data engineering. My PhD focused on designing and optimizing predictive models for prostate cancer using clinical and imaging data, with an emphasis on survival and outcome prediction.

EXPERIENCE

- **Qutwo** 🌐 06.2026 - Present
MTS (Artificial intelligence) Helsinki, Finland
 - Developing next-generation AI technologies for the quantum computing era.
 - Optimizing AI inference performance across GPU architectures.
- **University of Helsinki** 🌐 07.2026 - Present
Associated Researcher Helsinki, Finland
 - Continuing research in AI, machine learning, and computational pathology for prostate cancer.
- 01.2022 - 06.2026
Data Scientist Helsinki, Finland
 - Developed a data imputation algorithm to address missing time-series PCa data, improving EHR completeness by 25%.
 - Wrangled 300GB of Cancer data to conduct survival analysis, hypothesis testing, and machine learning training on 4000+ PCa patients.
 - Contributed to the development of HistoEncoder, a digital pathology foundation model for Prostate Cancer.
 - Presented technical solutions at international conferences and collaborated with international research teams.
- **Georgia Tech & Emory University** 🌐 04.2024 - 05.2024
Visiting Researcher (Machine Learning & Image Analysis) Atlanta, GA, USA
 - Engineered an image analysis pipeline to extract features from 40,000+ histopathology images (WSI), processing 20TB in 4 HPC servers and 16 GPUs.
 - Developed a multi-centre (FI, USA, JP) ML model to predict Prostate Cancer outcome, achieving 88% accuracy with an XGBoost after hyper-parameter optimisation using *Optuna*.
 - Optimized feature selection methods, including preparation-induced instability (PI) score and latent instability (LI) score, to quantify feature instability across and within datasets.
- **King Abdullah University of Science and Technology (KAUST)** 🌐 02.2024
Visiting Researcher (Statistical Modelling) Thuwal, Saudi Arabia
 - Built a Bayesian statistical model using INLAjoint R package to assess the Prostate Cancer treatment timing effect on future outcomes.
 - Developed a Bayesian model to predict metastatic cases in Prostate cancer using INLA methodology and INLAjoint package.
- **Aalto University** 🌐 02.2020 - 12.2021
Research Assistant (Machine Learning Engineering) Espoo, Finland
 - Developed a full-stack framework for real-time object detection from live video streaming, deployed on Nvidia Jetson GPUs.
 - Optimised edge-based parallel detection system, achieving 90% accuracy and reduced latency by 30%.
 - Contributed to a European project, focusing on scalable software solutions.

SKILLS

- **Languages:** English (Fluent), Arabic (Native), French (Fluent), Swedish (Elementary)
- **Programming:** Python, R, Bash, SQL, JavaScript, TypeScript
- **Data Science & AI/ML:** Supervised and Unsupervised Machine Learning, Deep learning, TensorFlow, PyTorch, Image processing, vLLM, Statistics, Time series, Hypothesis testing, Survival analysis
- **Cloud & Virtualisation:** Microsoft Azure, Azure ML, GCP, OpenStack-based cloud, Docker, Kubernetes
- **Web Technologies and Databases:** Django, NodeJS, REST APIs, MySQL, PostgreSQL, MongoDB
- **Other Tools:** Git, Parallel Programming (Ansible, Slurm), CI/CD Pipelines (Jenkins, Travis CI)

EDUCATION

• University of Helsinki <i>PhD in Data Science</i>	2022 - 2026 Helsinki, Finland
• Aalto University <i>Master's Exchange Studies in Data Science</i>	02.2020 - 09.2020 Espoo, Finland
• University of Science and Technology Houari Boumediene (USTHB) <i>Master's in Data Science</i>	09.2018 - 09.2020 Algiers, Algeria
• University of Science and Technology Houari Boumediene (USTHB) <i>Bachelor's in Computer Science and Mathematics</i>	09.2015 - 06.2018 Algiers, Algeria

PUBLICATIONS

A=ABSTRACT, C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [S.3] Batouche, et al. (202X). **Stability-Informed Multi-Institutional Classifier for Adverse Pathology Prediction in Radical Prostatectomy-Treated Prostate Cancer**. Under review. Preprint available upon request.
- [J.2] Batouche, et al. (2026). **A joint frailty model to assess the relationship between time to curative treatment and biochemical recurrence in prostate cancer patients**. In *Informatics in Medicine Unlocked*, 60, 101727, 2026. DOI: 10.1016/j.imu.2025.101727.
- [J.1] Batouche, et al. (2025). **Comparable recurrence risk for MRI-detected Gleason Grade Group (GG) 2 and systematic biopsy-detected GG1 prostate cancer**. In *BJU International*. DOI: 10.1111/bju.70109.
- [A.4] Batouche, et al. (2025). **Computational pathology-based classifier for predicting Gleason grade group up-grading on radical prostatectomy from diagnostic biopsies**. In the *Journal of Clinical Oncology*, 43, 329-329. 2025 ASCO Genitourinary Cancers Symposium, 13-15 February 2025, San Francisco, CA. DOI: 10.1200/JCO.2025.43-5_suppl.329.
- [S.2] Pohjonen, Batouche, et al. (2024). **HistoEncoder: a digital pathology foundation model for prostate cancer**. Under review. Preprint: DOI: 10.48550/arXiv.2411.11458.
- [A.3] Batouche, et al. (2024). **Joint modelling to assess the relationship between time to curative treatment and treatment recurrence in Prostate Cancer patients**. In *European Urology Open Science*, 63, S12-S13. 9th Baltic Meeting in conjunction with the EAU, 24-25 May 2024, Tallinn, Estonia. ISSN 2666-1683. DOI: 10.1016/S2666-1683(24)00424-5.
- [A.2] Batouche, et al. (2024). **Preliminary evidence of Gleason grade inflation in prostate cancer: A pre-treatment study**. In *European Urology Open Science*, 63, S14. 9th Baltic Meeting in conjunction with the EAU, 24-25 May 2024, Tallinn, Estonia. ISSN 2666-1683. DOI: 10.1016/S2666-1683(24)00425-7.
- [S.1] Batouche, et al. (2024). **MRI-Targeted Prostate Biopsy Introduces Grade Inflation and Overtreatment for Gleason Grade Group 2 cancers**. PMID: 38260466; PMCID: PMC10802666. DOI: 10.1101/2024.01.10.24300922. Updated version published in *BJU International* at DOI: 10.1111/bju.70109.
- [C.1] Batouche, et al. (2024). **Synergizing Data Imputation and Electronic Health Records for Advancing Prostate Cancer Research: Challenges, and Practical Applications**. In *Proceedings of the 17th International Joint Conference on Biomedical Engineering Systems and Technologies - HEALTHINF*, ISBN 978-989-758-688-0; ISSN 2184-4305, 77-86, SciTePress, 2024, Rome, Italy. DOI: 10.5220/0012350300003657
- [A.1] Batouche, et al. (2023). **Prognostic impact of prostate cancer grade inflation in targeted biopsies**. In *European Urology*, 83, S1371-S1372. Abstracts EAU23 - 38th Annual EAU Congress. ISSN 0302-2838. DOI: 10.1016/S0302-2838(23)00998-3.

GRANTS AND AWARDS

- **University Grant** 09.2024
University of Helsinki (Doctoral School)
◦ I have received a travel grant to participate in ASCO GU conference, San Francisco, CA, USA.
- **Best presentation award** 05.2024
European association of urology 
◦ I gave a talk at the 9th EAU Baltic meeting, Tallinn, Estonia, on our work focused on joint models
◦ The talk was followed by a poster presentation
◦ Our work was awarded the 2nd best presentation at the conference.
- **University Grant** 03.2024
University of Helsinki (Doctoral School)
◦ I have received a travel grant to participate in the International Joint Conference on Biomedical Engineering Systems and Technologies, Rome, Italy.
- **University Grant** 03.2023
University of Helsinki (Doctoral Programme in Computer Science)
◦ I have received a travel grant to attend the Oxford Machine Learning Summer School in Oxford, UK.

SUPERVISION AND TALKS

- **Supervision:** I co-supervised two Master's students in Computer Science at Aalto University, Finland, and serve as an Affiliated Supervisor at Lumiere Academy, guiding AI in health projects.
- **Conference presentations:** I presented my research at MENA ML 2026, ASCO 2025, ISCB 2024, EAU Baltic 2024, Health Informatics 2024, EAU 2023, ScanPath 2022, and other workshops and symposia.